# Cournot Model of Oligopoly

# Outlines

- Definition
- Assumptions

# Cournot Model

- The Cournot model, named after the French mathematician Augustin Cournot, is a basic economic model used to analyze competition between firms in an oligopoly market. In a Cournot duopoly model, there are only two firms competing against each other.

# Assumptions

- **Number of Firms:** There are two firms in the market, forming a duopoly. This allows for a simplified analysis of competition between a limited number of competitors.
- **Homogeneous Product:** Both firms produce the same product, with no differentiation in quality or features. Consumers perceive no difference between the products offered by the two firms.
- **Cost of Production:** The cost of production for each firm is zero. This simplifies the analysis by eliminating production costs as a factor in decision-making.

# Assumptions

- **Profit Maximization:** Each firm aims to maximize its profit. This involves determining the quantity of output that will generate the highest profit, taking into account the market demand and the competitor's output.
- **Demand Curve:** The demand curve is assumed to be linear, which means that the quantity demanded by consumers varies linearly with the price of the product.
- **Assumption of Fixed Output by Competitor:** Each firm assumes that its competitor will not change its output level. This allows each firm to make its production decision independently, without explicitly coordinating with its competitor.

# Assumptions

- **Decision-Making Process:** Each firm independently decides its output level to maximize its profit. However, firms take into account the likely response of their competitor when determining their own production levels.
- With these assumptions and characteristics, the Cournot model provides a framework for analyzing the strategic interaction between firms in a duopoly market, where each firm's output decision influences the market price and the competitor's profit-maximizing strategy.